

Project title

**FRAUD DETECTION IN FINANCIAL
TRANSACTIONS USING MICROSOFT POWER BI**

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ABSTRACT

Fraud remains a serious and developing concern in the financial sector, that can result in large financial losses, harm to one's reputation, and legal ramifications. Fraudulent activities have advanced beyond the capacity of conventional fraud detection systems due to the quick growth of digital banking, online transactions, and e-commerce. This project integrates business intelligence analytics and data visualization using Microsoft Power BI to improve fraud detection. A synthetic dataset has been created to mimic realistic financial transactions because real-world datasets are unavailable due to user privacy and data protection concerns. Key fraud detection methods, such as anomaly detection, pattern recognition, and predictive modelling, can be shown using this dataset. The system provides real-time insights and predictive alarms using interactive dashboards, DAX-based statistical analysis, and AI-powered visualizations, enabling stakeholders to take immediate action against possible dangers. Financial institutions may take pre-emptive measures to combat fraudulent activity thanks to Power BI's dynamic reporting, which improves transparency.

I. INTRODUCTION

In today's digital-first financial ecosystem, the rise of online banking, e-commerce, and instant payment platforms has brought both convenience and complexity. With this rapid evolution, fraudulent financial activities have also grown more sophisticated, posing serious challenges for institutions striving to safeguard their assets and reputation. Traditional fraud detection systems, often rule-based and reactive, struggle to keep pace with the dynamic and deceptive nature of modern fraud schemes. To address these limitations, financial organizations are increasingly turning to data-driven tools and intelligent systems. Among them, Microsoft Power BI has emerged as a powerful solution for turning complex transaction data into actionable insights.

II. DEFINITION

Fraud Detection in Financial Transactions refers to the systematic process of identifying and preventing unauthorized or suspicious financial activities, such as identity theft, unauthorized

transfers, and anomalous spending behaviour. This is achieved by analysing transaction data to uncover inconsistencies, unusual patterns, or behaviour that may indicate fraudulent intent.

III. CONCEPT

"Fraud Detection in Financial Transactions Using Microsoft Power BI" leverages real-time business intelligence to identify and prevent fraud. By transforming raw transaction data into interactive dashboards, Power BI enables anomaly detection through AI-driven analytics, DAX-based risk scoring, and visual tools and behaviour timelines. This streamlined approach helps businesses detect fraud faster, reduce risks, and make data-driven decisions with confidence.

IV. LITERATURE SURVEY

- **A. MACHINE LEARNING-BASED FRAUD DETECTION IN FINANCIAL TRANSACTIONS** This research examines supervised learning algorithms for fraud detection, including techniques like Decision Trees, Random Forest, and Support Vector Machines (SVM). The results indicate that Random Forest is the best performer, having high recall and precision in detecting fraudulent transactions.
- **B. DEEP LEARNING FOR CREDIT CARD FRAUD DETECTION** This work uses deep learning methods, namely Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), to model sequential transaction data. The work illustrates that deep learning models can identify underlying fraud patterns missed by conventional machine learning algorithms.
- **C. ANOMALY DETECTION IN TRANSACTIONS USING AI AND POWER BI** FINANCIAL This research explores the use of unsupervised machine learning algorithms like K-Means Clustering and Isolation Forest to detect fraudulent transactions. Combining AI-powered anomaly detection with Power BI allows for real-time fraud analysis via interactive dashboards.

V. COMPONENTS

- **A. DATA COLLECTION** The project's dataset has been synthetically generated to replicate actual financial transactions while protecting user anonymity. Key fields include TransactionID, UserID, TransactionAmount, TransactionType, MerchantCategory, TransactionDateTime, LocationCountry, and more.
- **B. DATA PREPROCESSING** Power BI's Power Query Advanced Editor was used for data pretreatment to guarantee data accuracy, consistency, and preparedness for

- analysis. Actions included handling data formatting, standardizing categorical variables, and geospatial processing.
- **C. EXPLORATORY DATA ANALYSIS (EDA)** DAX (Data Analysis Expressions) in Power BI was used to perform exploratory data analysis to find underlying transaction trends and possible fraud indications. The EDA focused on descriptive statistics, behavioural trends, outlier detection, and temporal patterns.
 - **D. DATA VISUALIZATION** The interactive dashboard features of Power BI were used to develop an extensive fraud detection interface, including dynamic dashboards, heatmaps, trend analysis charts, and geospatial maps.

VIII. ARCHITECTURE DIAGRAM

The given architecture diagram represents the workflow for analysing fraud detection in financial transactions using Microsoft Power BI. Here is the breakdown of the architecture.

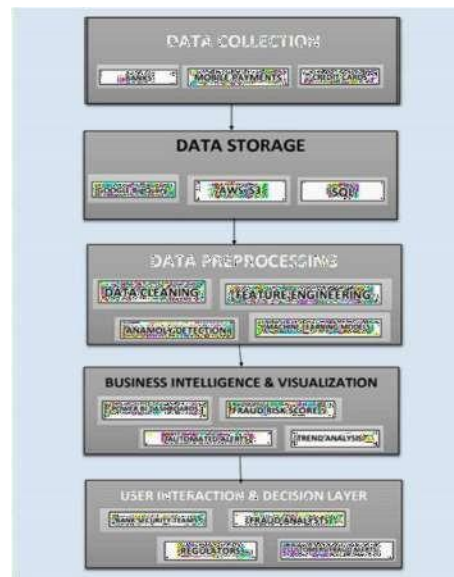


Fig. 1. Architecture Diagram

A. DATA COLLECTION LAYER The first step in the fraud detection process is the Data Collection Layer, where transactional data is generated synthetically.

B. DATA STORAGE LAYER The transaction data is stored in structured formats like Microsoft Excel or CSV files and imported into Power BI using Power Query.

C. DATA PREPROCESSING LAYER The raw data is cleaned, converted, and enhanced at the data preprocessing layer to prepare it for insightful analysis.

D. BUSINESS INTELLIGENCE & VISUALIZATION LAYER All prepared data is converted into usable insights using Power BI at the Business Intelligence & Visualization Layer, which consists of interactive dashboards, graphs, and reports.

RESULTS AND DISCUSSION

1. STACKED COLUMN CHART: High Transaction by Device

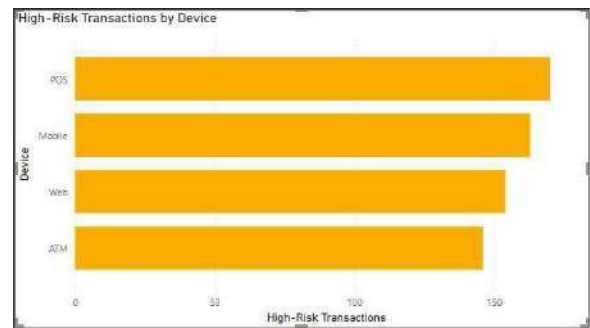


Fig. 2. High Transaction by Device

This bar chart shows the quantity of high-risk transactions across different device types: POS, mobile, web, and ATM. POS devices have the highest count, while ATMs have the lowest. This helps in prioritizing security efforts on the most vulnerable transaction routes.

2. KPI CARD: Total Transaction Amount and High-Risk Transaction



Fig. 3. Total Transaction Amount and High Rik Transaction

These graphic displays two key metrics: the total transaction amount (4 million) and the number of transactions marked as high-risk (633). These indicators offer a quick overview of the financial activity's scale and the associated fraud risk.

3. TABLE CHART: Fraud Risk Score Vs High Transaction Amounts

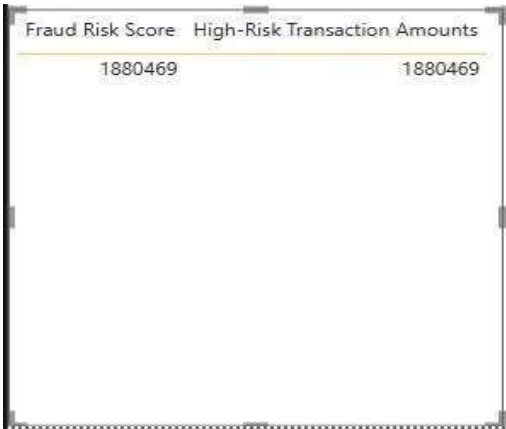


Fig. 4. Fraud Risk Score Vs High Transaction Amounts

This summary table shows that the cumulative Fraud Risk Score and the total High-Risk Transaction Amounts are both equal to 1,880,469. This highlights the direct financial significance of the transactions flagged as risky.

4. TABLE CHART: Ranking High Risk Transaction User

UserID	High-Risk Transactions	High-Risk Transaction Amounts	Fraud P
U001	9	35382	35382
U002	8	20772	20772
U003	7	11804	11804
U004	7	9139	9139
U005	8	37804	37804
U006	3	738	738
U007	3	658	658
U008	10	16331	16331
U009	8	18409	18409
U010	8	17858	17858
U011	9	17456	17456
U012	6	26890	26890
U013	6	15748	15748
U014	11	26284	26284
U015	10	18448	18448
U016	11	75117	75117
U017	9	35604	35604
U018	8	22743	22743
U019	8	46334	46334
U020	4	2772	2772
U021	10	25987	25987
U022	7	6091	6091
U023	10	24791	24791
U024	8	26111	26111
U025	2	1659	1659
U026	6	4310	4310
U027	2	9795	9795
U028	8	27284	27284
U029	8	14002	14002
U030	6	13028	13028
U031	4	10020	10020
Total	633	1880469	1880469

Fig. 5. Ranking High Risk Transaction User

This table provides a detailed analysis of high-risk transactions by user. Each row represents a single user, showing the number and value of their high-risk transactions. This is useful for identifying specific users who may require further investigation.

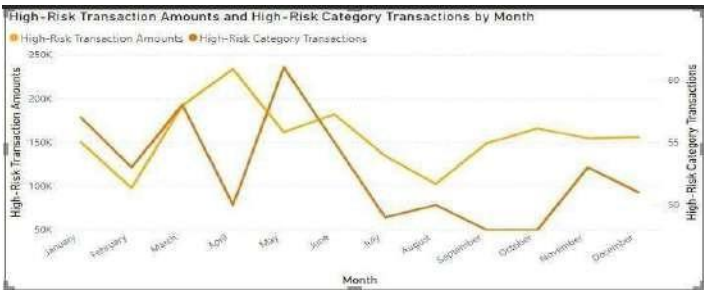


Fig. 6. Line chart on High-Risk Transaction And High Risk Transaction Category

The chart depicts monthly trends in the number and value of high-risk transactions. While the number of high-risk transactions remains relatively constant, there's a noticeable spike in the monetary value of these transactions in April and May.

6. STACKED COLUMN CHART: High Risk Transaction of users on an hourly basis

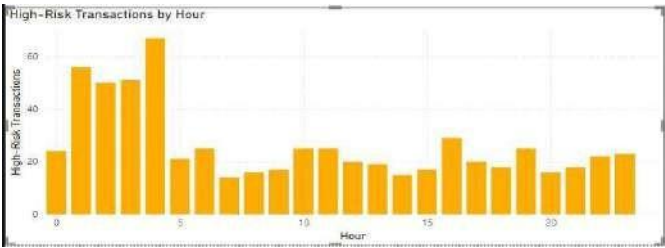


Fig. 7. High Risk Transaction of users on an hourly basis On the other hand, the quantity of high-risk transactions dramatically declines during the day, with comparatively steady and reduced activity noted starting at 6 AM. Fraud detection teams may find this pattern useful in more effectively allocating resources during periods of high risk.

This chart illustrates the frequency of high-risk transactions over a 24-hour period. The highest concentration of such transactions occurs between 1 AM and 5 AM, suggesting a pattern of fraudulent activity during late-night hours when monitoring may be less strict.

7. DONUT CHART: High Risk Transaction by Merchant Category

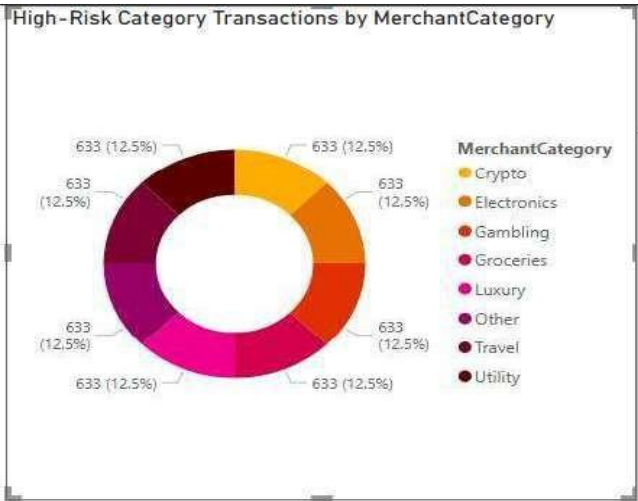


Fig. 8. Donut Chart representing High Risk Transaction by Merchant Category

This donut chart shows that high-risk transactions are equally distributed across all merchant categories, with each category (Crypto, Electronics, Gambling, etc.) accounting for 12.5% of the total.

8. PIE CHART: Risky Device Transactions

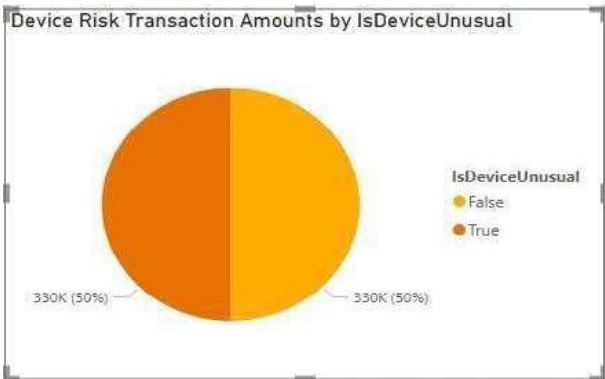


Fig. 9. Risky Device Transactions

Each of the two categories—"True" (unusual device) and "False" (usual device)—represents half of the 330K total transaction amount. This implies that both familiar and unfamiliar devices are equally likely to be the source of high-risk transaction amounts.

This graphic shows that the risk associated with transaction devices is split evenly, with 50% of the high-risk transaction amount coming from "usual" devices and 50% from "unusual" ones.

9. MAP CHART: Location based High Risk Transactions

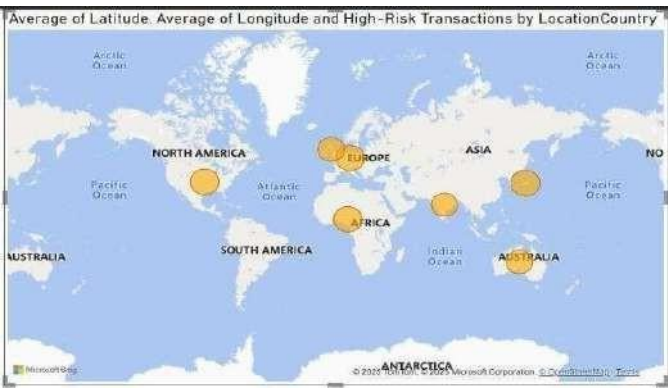


Fig. 10. Location based High Risk Transactions

This shows that high-risk actions are widespread around the world, with more frequent occurrences in some areas, underscoring the necessity of region-specific fraud surveillance.

This map displays the global distribution of high-risk transactions, with larger circles representing higher volumes. Noticeable concentrations are visible in North America, Europe, and Australia, highlighting the need for region-specific fraud monitoring.

10. AREA CHART: Categorizing High Risk Transaction on Hour

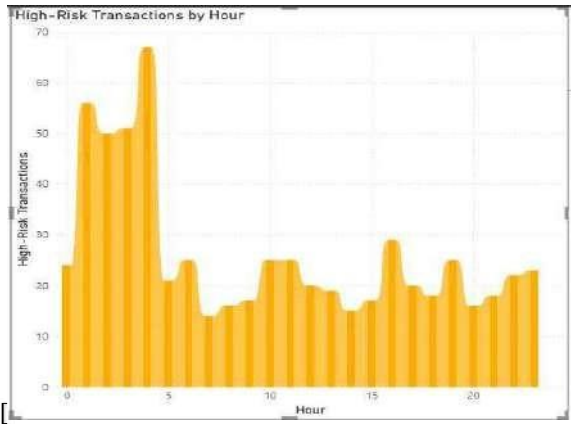


Fig. 11. Categorizing High Risk Transaction

This chart shows a noticeable spike in high-risk activity during the early morning hours, particularly between 12 AM and 5 AM, peaking at around 4 AM.

11. TABLE CHART: High Risk ATM Device Total Transaction Amount

Device	Device Risk Transactions	Device Risk Transaction Amounts
ATM	85	330358
Total	85	330358

Fig. 12. High Risk ATM Device Total Transaction Amount

The table focuses on high-risk transactions made via ATMs, showing a total of 85 such transactions amounting to a value of 330,358. This points to ATMs as a potential area for enhanced security.

12. MATRIX CHART: Hourly Total Risk Transaction Amount from 2021-2025

Year	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
2021	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100
2022	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100
2023	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100
2024	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100
2025	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100
Total	500	500	500	500	500	500	500	500	500	500	500	500	500	500	500	500	500	500	500	500	500	500	500	500

Fig. 13. Hourly Total Risk Transaction Amount from 2021-2025

This matrix displays high-risk transaction amounts by hour for the years 2021 to 2025. A consistent pattern of increased fraudulent activity is visible during the late-night hours (1 AM to 3 AM) across these years.

13. DASHBOARD 1: High-Risk Transaction Intelligence Dashboard



Fig. 14. High-Risk Transaction Intelligence Dashboard

This dashboard provides a comprehensive summary of high-risk transactions, focusing on user-level risk, transaction volume, and device usage. It effectively highlights user and device patterns in fraudulent activity.

14. DASHBOARD 2: Temporal & Behavioural Fraud Risk Dashboard

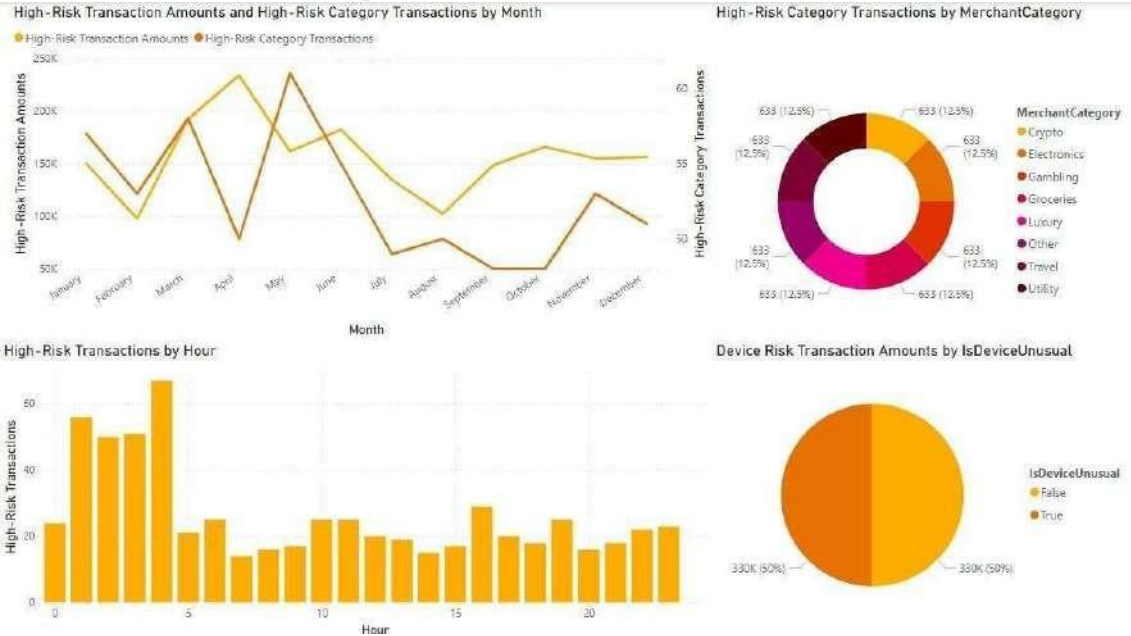


Fig. 15. Temporal & Behavioural Fraud Risk Dashboard

This dashboard offers a view of high-risk transactions through time-based, category-wise, and device-related analysis. It successfully identifies key risk areas related to timing, merchant categories, and device behaviour.

15. DASHBOARD 3: Hourly & Geographic Fraud Distribution Dashboard

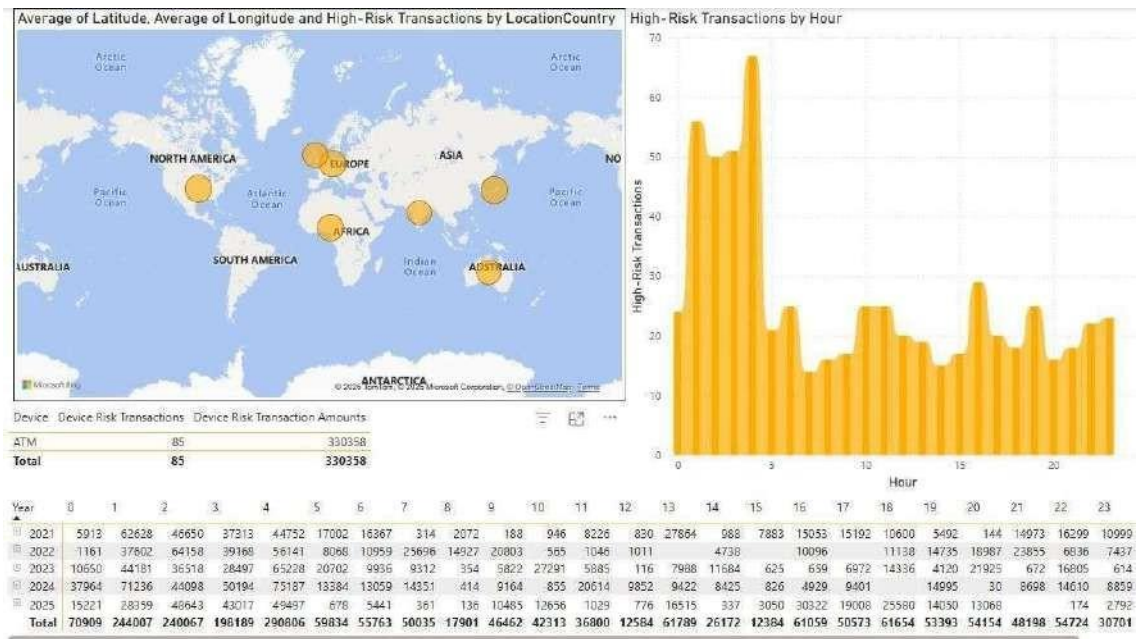


Fig. 16. Hourly & Geographic Fraud Distribution Dashboard

This dashboard provides a global perspective on high-risk transactions, with analytical patterns spanning time, geography, and device types. It helps stakeholders identify vulnerable times and locations.

CONCLUSION

This Power BI project provides an in-depth analysis of high-risk financial transactions, highlighting important trends across merchant categories, time, location, and device behaviour. The visualizations clearly demonstrate increases in fraudulent activity during specific months and early morning hours, indicating time-based vulnerabilities. The study emphasizes the need to monitor device abnormalities, as unusual device behaviour significantly increases transaction risk. By identifying key areas with high volumes of suspicious activity, geographic mapping enables location-specific interventions. Overall, the insights gained from this project can help improve fraud prevention techniques, facilitate better decision-making, and contribute to creating a more robust and secure financial ecosystem.

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